

Сборка ядра под ARM

1. Установка эмулятора, кросс-компилятора

```
pavel@pavel-MCLF-XX:~$ sudo apt install -y build-essential libncurses-dev bison flex libssl-dev libelf-dev bc qemu-system-arm gcc-arm-linux-gnueabi
```

2. Переход в папку с уже скачанным исходником ядра и чистка

```
pavel@pavel-MCLF-XX:~$ cd ~/PROG/KERNEL/linux
pavel@pavel-MCLF-XX:~/PROG/KERNEL/linux$ make mrproper
CLEAN arch/x86/boot/compressed
CLEAN arch/x86/boot
CLEAN arch/x86/entry/vdso
CLEAN arch/x86/kernel/cpu
CLEAN arch/x86/kernel
CLEAN arch/x86/realmode/rm
CLEAN arch/x86/tools
CLEAN arch/x86/lib
CLEAN certs
CLEAN drivers/firmware/efi/libstub
CLEAN drivers/scsi
CLEAN drivers/tty/vt
CLEAN init
CLEAN lib
CLEAN net/wireless
CLEAN security/selinux
CLEAN usr/include
CLEAN usr
CLEAN .
CLEAN modules.builtin modules.builtin.modinfo .vmlinux.export.c
CLEAN scripts/basic
CLEAN scripts/kconfig
CLEAN scripts/mod
CLEAN scripts/selinux/mdp
CLEAN scripts
CLEAN include/config include/generated arch/x86/include/generated debian .config .config.gdb .version Module.symvers
pavel@pavel-MCLF-XX:~/PROG/KERNEL/linux$
```

3. Сборка дефолтного конфига:

```
pavel@pavel-MCLF-XX:~/PROG/KERNEL/linux$ ARCH=arm make defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
HOSTCC scripts/kconfig/expr.o
LEX scripts/kconfig/lexer.lex.c
YACC scripts/kconfig/parser.tab.[ch]
HOSTCC scripts/kconfig/lexer.lex.o
HOSTCC scripts/kconfig/menu.o
HOSTCC scripts/kconfig/parser.tab.o
HOSTCC scripts/kconfig/preprocess.o
HOSTCC scripts/kconfig/symbol.o
HOSTCC scripts/kconfig/util.o
HOSTLD scripts/kconfig/conf
*** Default configuration is based on 'multi_v7_defconfig'
#
# configuration written to .config
#
pavel@pavel-MCLF-XX:~/PROG/KERNEL/linux$
```

.config - Linux/arm 6.14.0 Kernel Configuration

Linux/arm 6.14.0 Kernel Configuration

Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] built-in [] excluded <M> module < > module capable

General setup --->

(8) Maximum PAGE_SIZE order of alignment for DMA IOMMU buffers

System Type --->

Bus support --->

Kernel Features --->

Boot options --->

CPU Power Management --->

Floating point emulation --->

Power management options --->

General architecture-dependent options --->

[*] Enable loadable module support --->

[*] Enable the block layer --->

Executable file formats --->

Memory Management options --->

[*] Networking support --->

Device Drivers --->

File systems --->

Security options --->

-*- Cryptographic API --->

Library routines --->

Kernel hacking --->

4. Сама сборка и его расположение

```
pavel@pavel-MCLF-XX:~/PROG/KERNEL/linux$ ARCH=arm CROSS_COMPILE=arm-linux-gnueabihf- make -j12 zImage
SYNC    include/config/auto.conf.cmd
*
* Restart config...
*
```

